

Dynamic instability of microtubules as an efficient way to search in space

TIMOTHY E. HOLY AND STANISLAS LEIBLER

Departments of Physics and Molecular Biology, Princeton University, Princeton, NJ 08544

Communicated by Marc W. Kirschner, March 9, 1994

ABSTRACT During the formation of the mitotic spindle, dynamic microtubules search for chromosomes in the surrounding cytosol. The average time required to reach such “targets” is calculated within a simple model of dynamic instability. The values of the dynamic parameters that minimize this search time are also obtained. The results approximate well observations of the capture of chromosomes in prometaphase cells. Compared to usual reversible polymerization, dynamic instability reduces the time required to find a target by several orders of magnitude.

Microtubules are protein polymers that have a number of important biological functions in eukaryotic cells. They assemble by the process of dynamic instability (1), a process that differs substantially from the reversible polymerization of usual polymers (2). Reversible polymerization is a continuous competition between two processes, assembly and disassembly, where the monomer size controls the scale of the length fluctuations. In contrast, dynamic instability is, at the simplest level of description (3), a mechanism in which the polymer alternates between two distinct states of reversible polymerization. In one of these states the polymer grows; in the other it shrinks. One calls the transition from the growing to the shrinking state a catastrophe and the opposite transition a rescue. The average behavior of the polymer and the scale of the length fluctuations in dynamic instability depend strongly on the rates of switching between the two states. A microtubule can grow or shrink over distances of micrometers (thousands of monomers) before switching into the other state. Dynamic instability requires the continuous consumption of energy, which is provided by the hydrolysis of one molecule of GTP associated with each tubulin dimer (4).

It is believed that this unusual polymerization mechanism plays a crucial role in the biological functions of microtubules, but the precise nature of this role remains to be clarified. One interesting possibility, inspired by experiments (5–7) and investigated theoretically (8), is that dynamic instability is important in the regulation of tubulin synthesis. We reexamine and discuss this aspect elsewhere; here, we focus on the role of microtubules in morphogenesis. A model has been advanced (9) in which structure results from the interactions between microtubules and other objects in the cell. In this model, microtubules grow from a microtubule organizing center or a centrosome, “search” for a target object, and attach to it when they happen to come in contact. After attachment, a variety of responses may occur, such as stabilization of the connection or translocation of the object along the microtubule by molecular motors. This model of morphogenesis is supported, for example, by experiments on the capture of replicated chromosomes during mitosis (10, 11). An important feature of this picture is the random nature of the morphogenesis process; indeed, the location of the

target is not completely controlled before attachment to a microtubule and is influenced by Brownian motion and other randomizing forces.

In this paper, we address the question of the effectiveness of this “random morphogenesis process” by calculating the time required for a collection of polymers growing from a centrosome to find a target. (Note: We assume here that the cell seeks to decrease the search time to make the process more efficient. This time could of course be reduced if the cell increases the number of searching microtubules. However, since the synthesis of tubulin is energetically expensive, it is important for the cell to use its supply of monomers effectively.) We compute the average search time for two model polymers differing only in their polymerization mechanisms: a single-state polymer (SSP) that uses the mechanism of reversible polymerization (2) and a two-state polymer (TSP) that assembles through dynamic instability. SSPs evolve through a competition between assembly and disassembly; monomers are added at a rate r_{on} and are lost at a rate r_{off} . The SSP tip executes a biased random walk, with a step size δ equal to the monomer size. The TSP has two states, growing (g) and shrinking (s), in which the tip moves with constant velocity v_g and v_s , respectively. The polymer switches between these states stochastically, with frequencies of catastrophe (f_{cat}) and rescue (f_{res}) (12). For both polymer types we compute the average search time. Fig. 1*B Inset* illustrates the geometry of our system: polymers are nucleated from the center of a spherical cell of radius R , and they assemble and disassemble until they come into contact with a fixed target (see *Appendix A*) located a distance d from this center. We treat the polymers as rigid rods over distances of the order of the cell size (13), so the direction of growth is radial and fixed after nucleation (see *Appendix A*). For simplicity, we assume that the number of polymers is constant, so that when one polymer shrinks to zero length, another renucleates and grows in a new (randomly chosen) direction. (In real systems, the number of polymers fluctuates around some mean value, but the distribution will be sharply peaked when the number of polymers is large; thus assuming a constant number of polymers does not introduce significant error.) We choose to describe the cell membrane simply as a rigid wall that prevents further growth. (At the membrane an SSP reflects almost instantaneously, while a TSP stops growing but waits until it has a catastrophe before shrinking again.) While it should be obvious that cellular systems are much more complex, this model system is designed to illustrate the differences between the two polymer types as simply as possible, and the main conclusions should remain valid for more realistic situations.

The methods used to investigate this model are described in *Appendix B*.

RESULTS

The average search time depends on the geometric parameters and on the polymerization parameters (r_{on} and r_{off} for

Abbreviations: SSP, single-state polymer; TSP, two-state polymer.

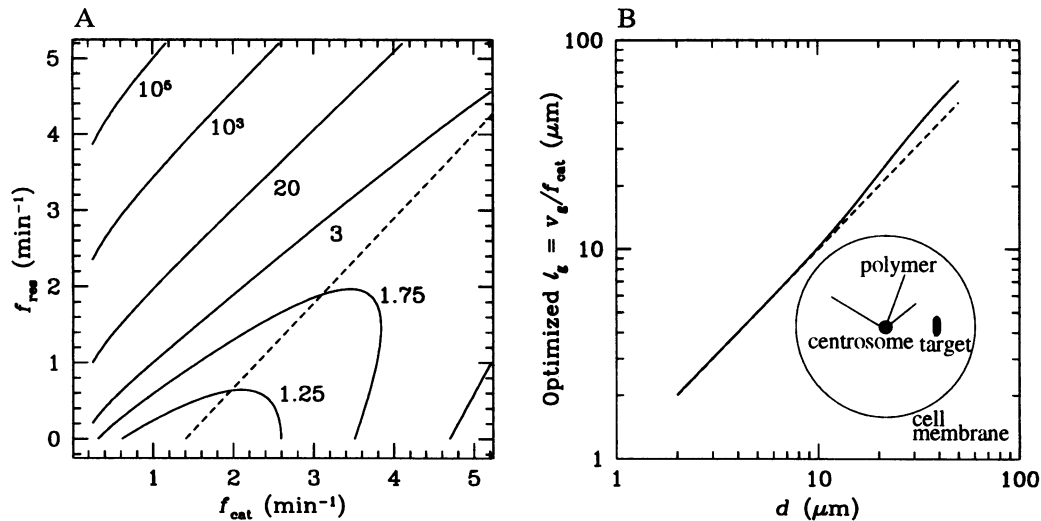


FIG. 1. Geometry of the simplified searching problem (*B Inset*) and dependence of the average search time τ_{TSP} on the TSP parameters f_{cat} and f_{res} . For a given distance d to the target, τ_{TSP} will have a minimum value τ_0 for a particular choice of f_{cat} and f_{res} . In *A* we display a contour plot of τ_{TSP}/τ_0 . Notice that the global minimum occurs for $f_{res} = 0$. The dashed line marks those values of f_{cat} and f_{res} for which the average TSP length ℓ is equal to d , where ℓ is given in terms of f_{cat} and f_{res} in Eq. 1. Notice that the dashed line lies close to the minimum trough in the contour plot. In *B* we show that, when $f_{res} = 0$, the optimal value of ℓ_g (solid line) is nearly equal to d (dashed line) over the entire range of d . ℓ_g is the average excursion size (equal to the average polymer length ℓ when $f_{res} = 0$) and is related to f_{cat} by $\ell_g = v_g/f_{cat}$. Here the TSP parameters are $v_g = 14.3 \mu\text{m}/\text{min}$, $v_s = 16 \mu\text{m}/\text{min}$ (11), the cell radius R is $50 \mu\text{m}$, and the target has a cross-sectional area of $1 \mu\text{m}^2$. In *A* the distance d to the target is $10 \mu\text{m}$, and in *B* we have set $f_{res} = 0$.

SSPs and v_g , v_s , f_{cat} , and f_{res} for TSPs). It is natural to ask which choices of the polymerization parameters minimize the search time. For SSPs searching for small targets, we find that the search time is minimized when $r_{on} \approx r_{off}$, so the evolution of the polymer is dominated by fluctuations rather than by drift. A collection of SSPs in a closed environment will automatically tend to the condition $r_{on} \approx r_{off}$ as the system approaches equilibrium (assembly depletes the supply of monomers, lowering r_{on} and making assembly less favorable). For TSPs, the behavior is somewhat more complex. In Fig. 1A, we show a contour plot of the search time as a function of both f_{cat} and f_{res} . The detailed calculation shows that the search time is shortest when the average length ℓ of the polymers is close to d , the distance to the targets. ℓ depends on the catastrophe and rescue rates (14) as:

$$\ell = \frac{v_g v_s}{f_{cat} v_s - f_{res} v_g}, \quad [1]$$

so $\ell \approx d$ can be written as a relation between f_{cat} and f_{res} . The locus defined by $\ell = d$ lies close to the minimum trough in the contour plot (Fig. 1A). The global minimum is achieved for $\ell \approx d$ and $f_{res} = 0$. When the rescue rate f_{res} is zero, Eq. 1 shows that ℓ is equal to $\ell_g = v_g/f_{cat}$, where ℓ_g can be interpreted as the average distance moved by the tip while in the growing state. Thus the global minimum condition is $f_{res} = 0$ and $\ell_g \approx d$ (see Fig. 1B). Note from Fig. 1A that small deviations from this global minimum condition do not significantly change the search time as long as $\ell \approx d$. However, if this condition does not hold, the search time can increase dramatically.

In Fig. 2, we choose fixed SSP and TSP parameters and show how the average search time depends on the target distance d . When d is not much larger than ℓ_g , TSPs find targets faster than SSPs by a large factor, which for typical values of the parameters may be as big as 10^3 . The detailed analysis (see *Appendix B*) shows that the dominant contribution to this factor is the ratio of the cell size R to the monomer size δ . Dynamic instability amplifies the spatial fluctuations in the polymer length and changes their characteristic scale from δ to $\ell_g \approx d \sim R$. These large fluctuations

give dynamically unstable polymers a clear advantage in finding targets.

Fig. 2 also clearly shows that τ_{TSP} increases very rapidly as d/ℓ_g becomes much larger than unity; this happens because TSPs rarely grow much longer than ℓ_g . This means that if a target is displaced further than the typical target distance for which the parameters were optimized, the search becomes much (exponentially) slower.

DISCUSSION

It is interesting to compare these results to observations in living systems. For concreteness, we use results from newt

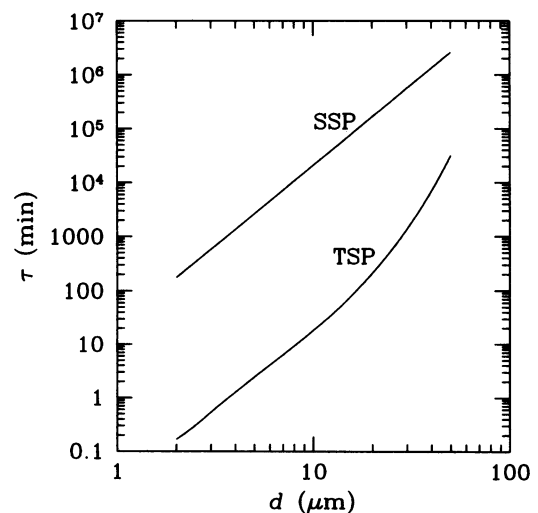


FIG. 2. Average search time as a function of the distance d to the target for a search by 250 polymers. Note that the times for SSPs are roughly 10^3 times larger than the TSP times over much of the range of d . Here we have fixed $f_{cat} = 1.52 \text{ min}^{-1}$, and the other geometric and TSP parameters are as in Fig. 1B (yielding $\ell_g = 9.41 \mu\text{m}$). The SSP step size δ is 8 nm, and the rates are $r_{on} = r_{off} = (v_g + v_s)/2\delta \approx 1890 \text{ min}^{-1}$, so the monomer reaction rates are the same on average for the two polymer types.

lung pneumocytes, a system that has been the focus of a number of experiments (10, 11, 15, 16). (i) Note that the optimized TSP search parameters $\ell_g \approx d$ and $f_{\text{res}} = 0$ found above seem to be remarkably close to microtubule dynamic parameters during prometaphase, if d is taken as the typical distance from the spindle poles to the chromosomes [typically $\ell_g \approx 10 \mu\text{m}$ and $d = 10\text{--}20 \mu\text{m}$ (11)]. (ii) It has been determined that the time required for microtubules to capture the kinetochores in mitotic newt lung pneumocytes is usually on the order of 10 min, under usual circumstances, when the chromosomes are closer than $30 \mu\text{m}$. In a few cases, the chromosomes become displaced from either spindle pole by as much as $50 \mu\text{m}$, and the time required to capture the chromosome can be more than 5 h (10). These times show the same strong distance dependence displayed by τ_{TSP} in Fig. 2, and in fact these times match experimental values in order of magnitude. Although one should keep in mind the additional phenomena that alter the process of target finding (diffusion of chromosomes, inhomogeneities of the cytosol, bending of microtubules, motion of the asters, etc.), these results argue in support of the relevance of our analysis.

Our calculation can be extended in a number of ways. The diffusion of targets, the effects of finite microtubule persistence length, and the possibility of multiple targets distributed over a range of distances from the centrosome can be investigated. However, even at this point, the present model clearly demonstrates the importance of dynamic instability in the search for objects within the cell. Further quantitative experiments on the capture of chromosomes during prometaphase would be important in verifying our results. In particular, it would be interesting to check whether and how the cell solves the problem of exponentially increasing searching times when $d > \ell_g$. One possible solution is the existence of an additional biochemical control mechanism(s) that could "guide" the microtubules toward such "lost" targets.

APPENDIX A

It is necessary to consider the Brownian motion of the targets if one wishes to apply the analysis to a number of biologically relevant systems, e.g., the capture of chromosomes. Since chromosomes can attach laterally to preexisting polymers (17), we need to determine the amount of time t_D for a chromosome to diffuse the distance between polymers and compare it to the lifetime of a polymer of length d . It is easy to see that the separation between N radially growing polymers at a distance d from the center is on the order of $d\sqrt{4\pi/N}$. The time required to diffuse this distance is $t_D = 4\pi d^2/ND$, where D is the diffusion constant of the chromosome. Realistic values of the parameters are $D \approx 0.01 \mu\text{m}^2/\text{min}$ [as calculated from viscosity measurements (10)], $d = 10 \mu\text{m}$, and $N = 250$ in newt lung pneumocytes, yielding $t_D \approx 10^3 \text{ min}$. This is to be compared to the average lifetime of a polymer of length d , which is on the order of $d/v \approx 1 \text{ min}$ for TSPs and $Rd/k\delta^2 \approx 10^3 \text{ min}$ for SSPs (where $v \approx v_{g/s}$, $r \sim r_{\text{on/off}}$, and the numerical values are chosen as in Fig. 2). Thus we expect chromosome diffusion to be unimportant for TSPs and important but not dominant for SSPs. This estimate also shows that the ratio of the search times $\tau_{\text{SSP}}/\tau_{\text{TSP}}$ calculated below may be reduced at most by an order of magnitude from this effect.

Another important issue is whether the polymer tips themselves can move laterally, either through bending of the polymer or by pivoting on the attachment site. Observations of microtubules in *Xenopus* extracts indicate that the attachment site is not a free pivot; we estimate that for a $10\text{-}\mu\text{m}$ microtubule, the tip is able to move less than $1 \mu\text{m}$ laterally. This motion is rapid compared to the times of catastrophes and rescues; therefore, it effectively increases the cross

section of the interaction between the microtubule and the target. In our calculation of the search time for kinetochores, we chose the cross section to be large enough to include these effects.

APPENDIX B

The details of the calculation will be presented elsewhere. Here we describe the main points. The entire search is composed of two types of primitive events: "unsuccessful" searches, in which a polymer grows and then depolymerizes completely before finding the target, and "successful" searches, in which a polymer grows and finds the target before it has a chance to depolymerize completely. Polymers make unsuccessful searches and are replaced by new polymers, which in turn make further unsuccessful searches; the process ends when one polymer makes a successful search. The time for the entire search is calculated in two steps: (i) we compute the probabilities and average durations of both successful and unsuccessful searches; (ii) we sum the probabilities and durations over all possible sequences of these searching events. The analytical tool for the first step is the first-passage time distribution (18), computed in the presence of absorbing and/or reflecting walls (the centrosome and cell membrane). The probabilities and durations of the primitive search events are the lowest moments of the first-passage time distribution. We can calculate the moments of these distributions analytically. The sum over all sequences of searching events can also be computed analytically when the number of polymers is one. For multiple polymers, the total search time depends on higher moments, and numerical calculation and/or analytical approximations are necessary.

We illustrate our method with a simple heuristic estimate of the search time for a single polymer. Let p and t_s be the probability and average duration, respectively, of a successful search and let t_u be the average duration of an unsuccessful search. Then, the average search time τ can be written as a sum over all possible sequences of unsuccessful and successful searching events,

$$\begin{aligned}\tau &= pt_s + (1-p)p(t_s + t_u) + (1-p)^2p(t_s + 2t_u) + \dots \\ &= t_s + \frac{1-p}{p} t_u.\end{aligned}\quad [\text{B.1}]$$

If the target is small, then $p \ll 1$, so $\tau \approx t_u/p$. Thus we only need to estimate p and t_u . To compute p , notice that a successful search requires the conjunction of two events: the polymer has to grow in the direction of the target and to become long enough to reach the target before it depolymerizes completely. Thus we write $p = qp^*$, where q is the probability of growing in the direction of the target (proportional to the angular size of the target) and p^* is the probability of reaching length d before depolymerizing completely. For a TSP, p^* is clearly e^{-d/ℓ_s} when $f_{\text{res}} = 0$. For a SSP, p^* is δ/d when $r_{\text{on}} = r_{\text{off}} = r$. Thus, the average number of unsuccessful searches is $e^{d/\ell_s}/q$ for a TSP and $d/\delta q$ for a SSP. The duration of an unsuccessful search follows similarly: it is on the order of ℓ_g/v for a TSP (here $v_g = v_s = v$) and $R/r\delta$ for a SSP. Putting everything together, we obtain

$$\tau_{\text{TSP}} \approx \frac{\ell_g e^{d/\ell_s}}{qv} \quad \text{and} \quad \tau_{\text{SSP}} \approx \frac{Rd}{qr\delta^2}.\quad [\text{B.2}]$$

We see that the minimum of τ_{TSP} is achieved for $\ell_g = d$. If we put $v = r\delta$ (so the assembly and disassembly rates are the same for the two polymer types) and set $\ell_g \approx d$, the ratio of these two times is

$$\frac{\tau_{SSP}}{\tau_{TSP}} \approx \frac{Rd}{\delta \ell_g} e^{-d/\ell_g} \approx \frac{R}{\delta}, \quad [\text{B.3}]$$

as claimed in *Results*.

Note Added in Proof. Experiments on the capture of chromosomes during mitosis in rat kangaroo cells (19) also provide evidence that capture time depends on distance in a way that is consistent with our calculation. We thank Conly Rieder for directing our attention to this reference.

We thank M. Dogterom, E. Karsenti, M. Kirschner, T. Mitchison, M. Rose, and J. Stock for valuable discussions and comments. T.E.H. is supported by a National Science Foundation Fellowship.

1. Mitchison, T. & Kirschner, M. (1984) *Nature (London)* **312**, 237–242.
2. Oosawa, F. & Asakura, S. (1975) *Thermodynamics of the Polymerization of Proteins* (Academic, New York).
3. Hill, T. L., ed. (1987) *Linear Aggregation Theory in Cell Biology*, Springer Series in Molecular Biology (Springer, New York).
4. Caplow, M. (1992) *Curr. Opin. Cell Biol.* **4**, 58–65.
5. Ben Ze'ev, A., Farmer, S. R. & Penman, S. (1979) *Cell* **17**, 319–325.
6. Cleveland, D. W., Lopata, M. A., Sherline, P. & Kirschner, M. W. (1981) *Cell* **25**, 537–546.
7. Caron, J. M., Jones, A. L., Rall, L. B. & Kirschner, M. W. (1985) *Nature (London)* **317**, 648–649.
8. Mitchison, T. J. & Kirschner, M. (1987) *Cell Biophys.* **11**, 35–55.
9. Kirschner, M. & Mitchison, T. (1986) *Cell* **45**, 329–342.
10. Rieder, C. L. & Alexander, S. P. (1990) *J. Cell Biol.* **110**, 81–95.
11. Hayden, J. H., Bowser, S. S. & Rieder, C. L. (1990) *J. Cell Biol.* **111**, 1039–1045.
12. Gildersleeve, R. F., Cross, A. R., Cullen, K. E., Fagen, A. P. & Williams, R. C., Jr. (1992) *J. Biol. Chem.* **267**, 7995–8006.
13. Gittes, F., Mickey, B., Nettleton, J. & Howard, J. (1993) *J. Cell Biol.* **120**, 923–934.
14. Verde, F., Dogterom, M., Stelzer, E., Karsenti, E. & Leibler, S. (1992) *J. Cell Biol.* **118**, 1097–1108.
15. Cassimeris, L., Pryer, N. K. & Salmon, E. D. (1988) *J. Cell Biol.* **107**, 2223–2231.
16. Alexander, S. P. & Rieder, C. L. (1991) *J. Cell Biol.* **113**, 805–815.
17. McIntosh, J. R. & Hering, G. E. (1991) *Int. Rev. Cell Biol.* **7**, 403–426.
18. van Kampen, N. G. (1992) *Stochastic Processes in Physics and Chemistry* (North-Holland, Amsterdam, The Netherlands), 2nd Ed.
19. Roos, U.-P. (1976) *Chromosome* **54**, 363–385.